

ASU ROAR

Continuous Global Planner Performance & Path Geometry Evaluation

OVERALL SYSTEM EVALUATION: SYSTEM STABLE / PERFORMANCE ENVELOPE MAINTAINED

System Summary Diagnostics

PATH SUCCESS RATE	SIGNAL SMOOTHNESS	MAX TURN ANGLE	MIN OBSTACLE CLEARANCE
100.0%	0.1949 s	45.0°	0.00 m

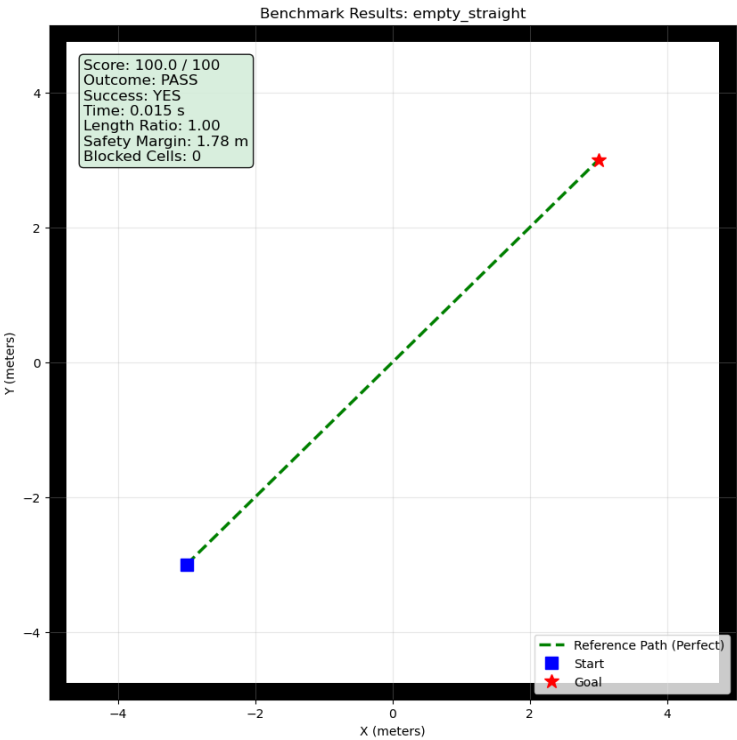
Scenario Performance Breakdown

Scenario ID	Status	Planning Time	Distance	Ratio	Max Turn	Min Clearance	Cost
empty_straight	PASS	0.014 s	8.49 m	1.00	0.0°	1.78 m	0.0
scattered_rocks_detour	FAIL	0.171 s	11.64 m	1.03	45.0°	0.04 m	0.0
canyon_gate_passage	PASS	0.017 s	8.00 m	1.00	0.0°	0.78 m	0.0
marsyard_rough_slopes	PASS	0.516 s	13.66 m	1.21	45.0°	0.73 m	0.0
marsyard_labyrinth	FAIL	0.476 s	20.16 m	1.78	45.0°	0.04 m	0.0
canyon_gate_blocked	PASS	0.017 s	8.00 m	1.00	0.0°	0.78 m	0.0
marsyard_snake_passage	FAIL	0.259 s	11.96 m	1.06	45.0°	0.04 m	0.0
dead_end_trap	FAIL	0.006 s	5.00 m	1.00	0.0°	0.00 m	300.0
crater_field	PASS	0.278 s	12.19 m	1.08	45.0°	0.78 m	0.0

Detailed Diagnostics: empty_straight

Metric	Measured Value
Path Compilation Time	0.0144 s
Distance Covered (Path Length)	8.49 m
Path Length / Euclidean Ratio	1.000
Turn Angles	Min: 0.0° Max: 0.0° Avg: 0.0°
Turn Radii	Min: 0.00 m Max: 0.00 m Avg: 0.00 m
Nearest Distance to Obstacle (Clearance)	1.78 m
Total Path Cost	0.0

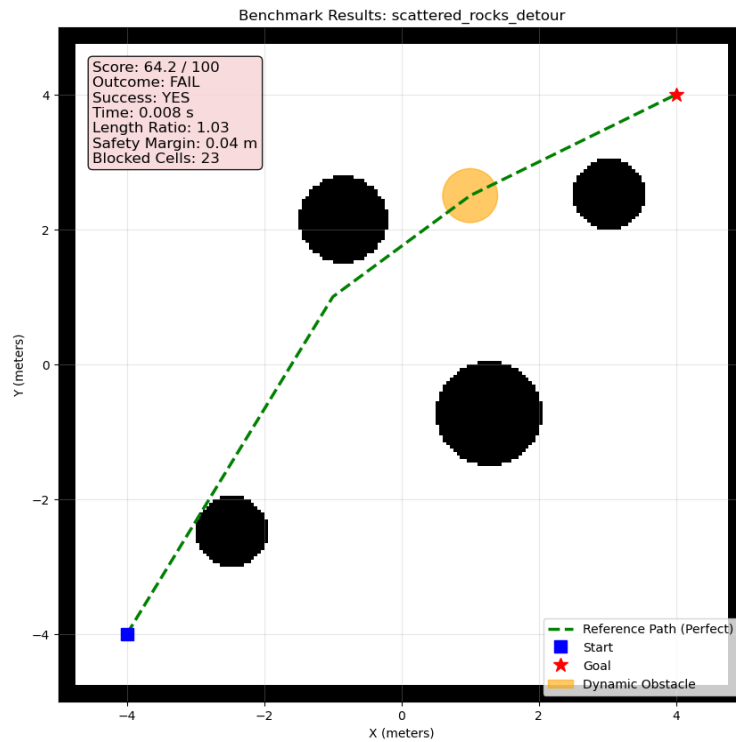
Path Visualization



Detailed Diagnostics: scattered_rocks_detour

Metric	Measured Value
Path Compilation Time	0.1709 s
Distance Covered (Path Length)	11.64 m
Path Length / Euclidean Ratio	1.028
Turn Angles	Min: 0.0° Max: 45.0° Avg: 5.6°
Turn Radii	Min: 0.08 m Max: 0.08 m Avg: 0.08 m
Nearest Distance to Obstacle (Clearance)	0.04 m
Total Path Cost	0.0

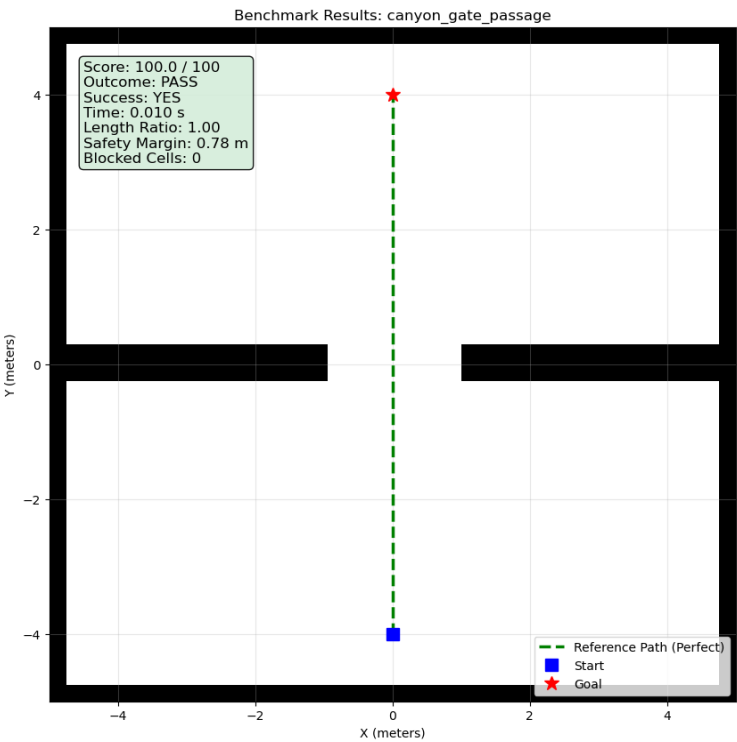
Path Visualization



Detailed Diagnostics: canyon_gate_passage

Metric	Measured Value
Path Compilation Time	0.0168 s
Distance Covered (Path Length)	8.00 m
Path Length / Euclidean Ratio	1.000
Turn Angles	Min: 0.0° Max: 0.0° Avg: 0.0°
Turn Radii	Min: 0.00 m Max: 0.00 m Avg: 0.00 m
Nearest Distance to Obstacle (Clearance)	0.78 m
Total Path Cost	0.0

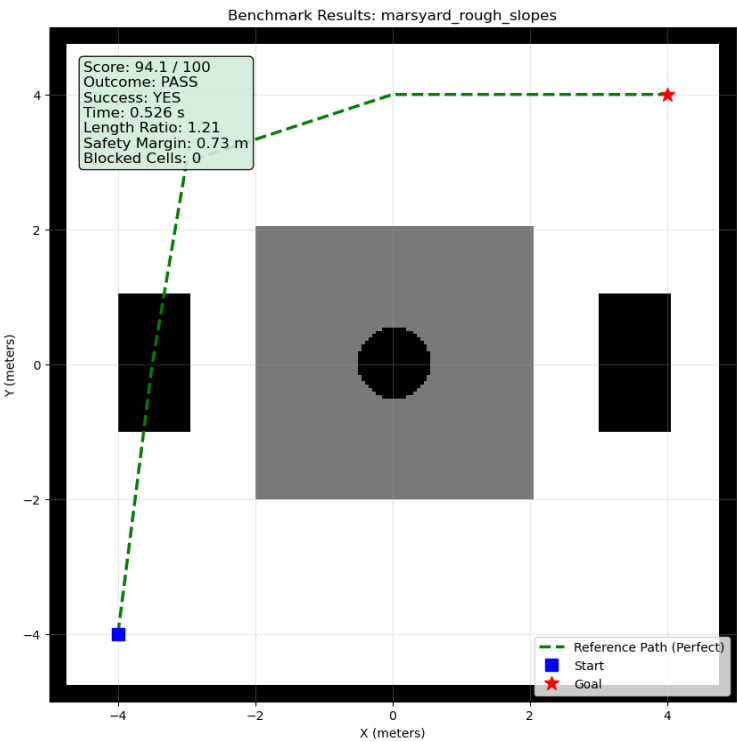
Path Visualization



Detailed Diagnostics: marsyard_rough_slopes

Metric	Measured Value
Path Compilation Time	0.5164 s
Distance Covered (Path Length)	13.66 m
Path Length / Euclidean Ratio	1.207
Turn Angles	Min: 0.0° Max: 45.0° Avg: 16.0°
Turn Radii	Min: 0.08 m Max: 0.08 m Avg: 0.08 m
Nearest Distance to Obstacle (Clearance)	0.73 m
Total Path Cost	0.0

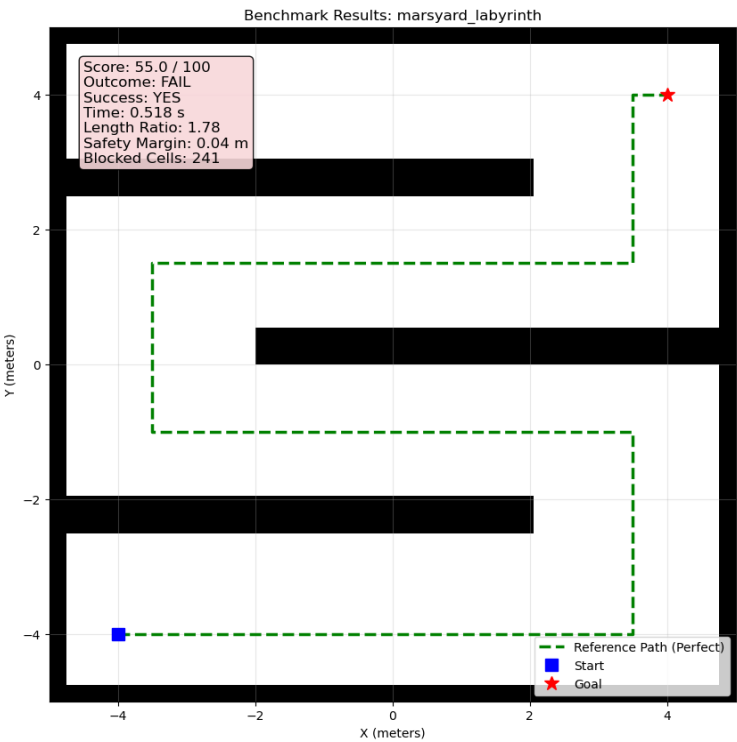
Path Visualization



Detailed Diagnostics: marsyard_labyrinth

Metric	Measured Value
Path Compilation Time	0.4760 s
Distance Covered (Path Length)	20.16 m
Path Length / Euclidean Ratio	1.782
Turn Angles	Min: 0.0° Max: 45.0° Avg: 2.2°
Turn Radii	Min: 0.08 m Max: 0.08 m Avg: 0.08 m
Nearest Distance to Obstacle (Clearance)	0.04 m
Total Path Cost	0.0

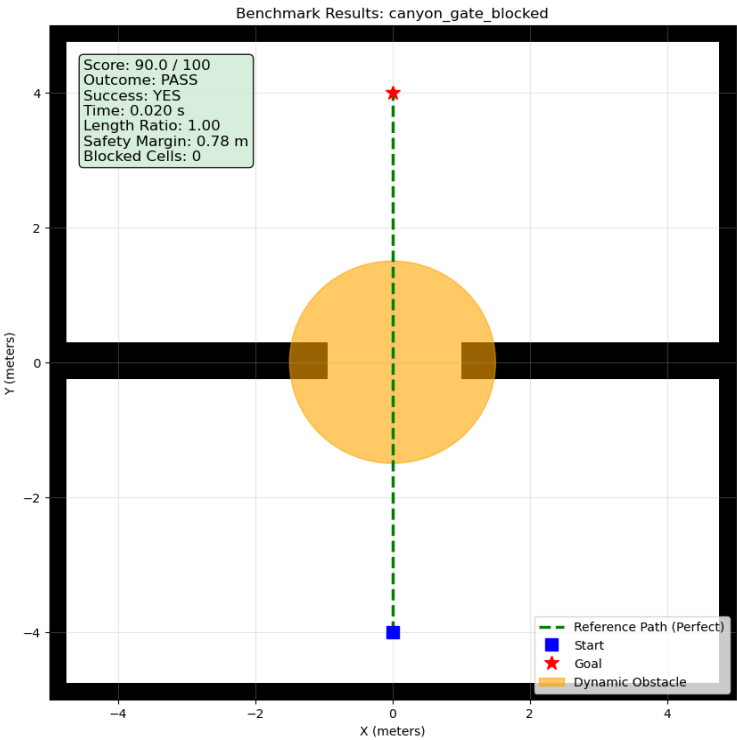
Path Visualization



Detailed Diagnostics: canyon_gate_blocked

Metric	Measured Value
Path Compilation Time	0.0168 s
Distance Covered (Path Length)	8.00 m
Path Length / Euclidean Ratio	1.000
Turn Angles	Min: 0.0° Max: 0.0° Avg: 0.0°
Turn Radii	Min: 0.00 m Max: 0.00 m Avg: 0.00 m
Nearest Distance to Obstacle (Clearance)	0.78 m
Total Path Cost	0.0

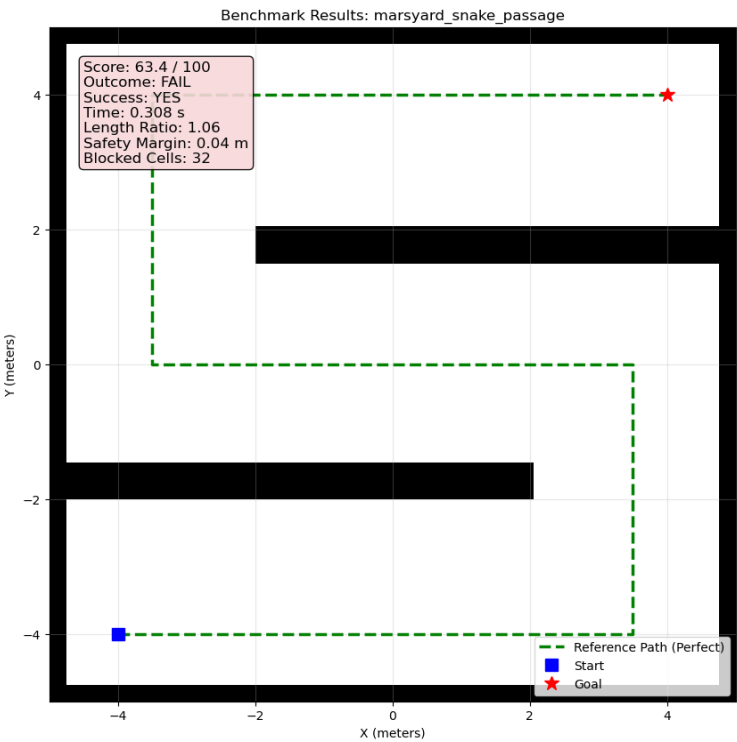
Path Visualization



Detailed Diagnostics: marsyard_snake_passage

Metric	Measured Value
Path Compilation Time	0.2589 s
Distance Covered (Path Length)	11.96 m
Path Length / Euclidean Ratio	1.057
Turn Angles	Min: 0.0° Max: 45.0° Avg: 6.7°
Turn Radii	Min: 0.08 m Max: 0.08 m Avg: 0.08 m
Nearest Distance to Obstacle (Clearance)	0.04 m
Total Path Cost	0.0

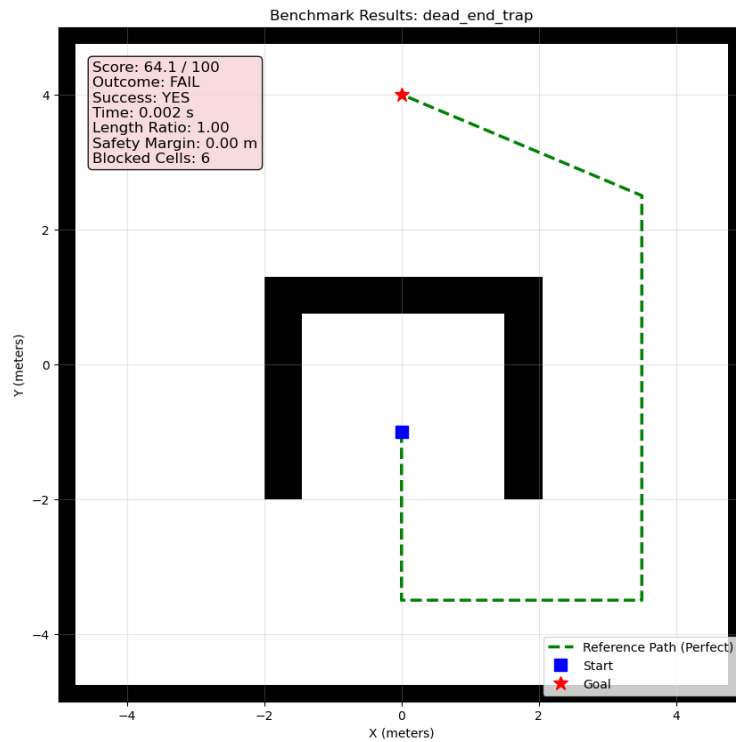
Path Visualization



Detailed Diagnostics: dead_end_trap

Metric	Measured Value
Path Compilation Time	0.0062 s
Distance Covered (Path Length)	5.00 m
Path Length / Euclidean Ratio	1.000
Turn Angles	Min: 0.0° Max: 0.0° Avg: 0.0°
Turn Radii	Min: 0.00 m Max: 0.00 m Avg: 0.00 m
Nearest Distance to Obstacle (Clearance)	0.00 m
Total Path Cost	300.0

Path Visualization



Detailed Diagnostics: crater_field

Metric	Measured Value
Path Compilation Time	0.2775 s
Distance Covered (Path Length)	12.19 m
Path Length / Euclidean Ratio	1.078
Turn Angles	Min: 0.0° Max: 45.0° Avg: 16.9°
Turn Radii	Min: 0.08 m Max: 0.08 m Avg: 0.08 m
Nearest Distance to Obstacle (Clearance)	0.78 m
Total Path Cost	0.0

Path Visualization

